

Let the Sheet Do the Work — Paper C

HARD • NO COMPUTER • NO CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. There is no computer and no calculator: every number is designed to work by hand.
- This paper is about **Google Sheets**. Every formula starts with **=**, arguments are separated by **commas**, and text goes in **quotation marks**.
- Where a question asks for a **value**, write what the cell would show. Where it asks for a **formula**, write it exactly, brackets and commas included.
- The mark for each question is shown in square brackets on the right.
- No list of functions is printed on this paper. You are expected to remember the syntax, including the order of the arguments.
- Several questions describe sheets you may not have seen before. You are not expected to recognise them: apply the ideas you already have.
- A correct value with a wrong formula, or the other way round, earns one mark and not two.

SECTION 1 – READ THE SHEET

- 1.** A school library records how many days late each book came back. Column C works out the fine at ₹2 a day. Chirag's book was lost, so the librarian typed the word **lost**; Disha's cell was left empty. The extract shows column C as typed.

	A	B	C
1	Student	Days late	Fine (₹)
2	Akash	3	=B2*2
3	Bhumi	7	=B3*2
4	Chirag	lost	=B4*2
5	Disha		=B5*2
6	Eklavya	5	=B6*2
7	Fiza	9	=B7*2

(a) What does cell C4 show? [1]

(b) State the values of =COUNT(B2:B7) and of =COUNTA(B2:B7). [1]

(c) State the value of =SUM(B2:B7). [1]

(d) State the value of =AVERAGE(B2:B7). [1]

(e) State the value of =SUM(B2:B4), and explain why it is not an error although C4 is. [1]

2. A cricket sheet lists eight players, with Team in column B and Runs in column C.

	A	B	C
1	Player	Team	Runs
2	Aryan	Nilgiri	62
3	Vivaan	Nilgiri	45
4	Krish	Nilgiri	78
5	Dhruv	Aravalli	91
6	Atharv	Aravalli	33
7	Yuvraj	Aravalli	50
8	Rudra	Sahyadri	24
9	Shaurya	Sahyadri	70

(a) A filter is applied: **Team is exactly “Aravalli”** and **Runs is greater than or equal to 50**. Write the row numbers that remain visible. [2]

(b) The filter is removed. The **whole table** is sorted by Runs, largest first. Write all eight names in order, top to bottom. [2]

(c) The sort is undone and the filter from (a) is switched back on. A student types =SUM(C2:C9) in an empty cell below the table. State the value it shows. [1]

3. A coach adds a Strike rate column: the runs a batter scores per 100 balls. Runs are in column C and balls faced in column D, rows 2 to 8. The Team is in column B.

	A	B	C	D	E
1	Player	Team	Runs	Balls	Strike rate
2	Aryan	Nilgiri	80	50	
3	Vivaan	Nilgiri	55	50	
4	Krish	Nilgiri	75	50	
5	Dhruv	Aravalli	40	50	
6	Atharv	Aravalli	50	40	
7	Rudra	Sahyadri	24	30	
8	Shaurya	Sahyadri	30	20	

- (a) Write the formula for E2, so that it can be filled down. [1]

- (b) Write one formula that counts the players who scored **50 or more**. [1]

- (c) Write one formula that gives the average runs of the **Nilgiri** players only, and state its value. [2]

4. A river-gauge sheet records the level of a river in whole metres. The level is typed in A2. Column B should flag it:

12 m or more: Danger · 9 to 11 m: Alert · 5 to 8 m: Watch · below 5 m: Normal

- (a) Write the formula for B2 using nested IF. [3]

- (b) State what your formula shows for a level of 12, 9, 8 and 4. [2]

- (c) A different formula in B2 begins [1]
=IF(A2>=5, "Watch", IF(A2>=9, "Alert", IF(A2>=12, "Danger", "Normal"))). State what it shows for a level of 14 m, and say why.
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SECTION 3 – DIAGNOSE AND DESIGN

5. A feedback form records a star rating from 1 to 5 in column B.

- (a) Design a data validation rule for column B. State the criteria with the numbers, [2]
and the setting for invalid data.
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- (b) With your rule in place, someone types each of these in turn: 0, 5, 6 and the [1]
word five. State which are accepted.
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- (c) Last week, before the rule existed, the cells B9 and B10 were filled with 9 and [1]
with five. State what happens to those two cells when the rule is added, and what the owner must do.
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6. A tuck shop wants to decide what to stock. Part of its sheet is shown. Column D is meant to hold the revenue: price times number sold.

	A	B	C	D
1	Item	Price (₹)	Sold	Revenue (₹)
2	Chai	10	90	
3	Samosa	15	70	
4	Poha	30	45	
5	Thali	80	20	

- (a) Write the revenue of each item in column D. Then say which item sold the most [2]
units and which item earned the most **money**.
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- (b) Write the formula that gives the total revenue of the four items, and state its value. [1]

- (c) The shop can stock only two of the four items. Which two would you choose if the aim is to earn the most money? Which two if the aim is to serve the most children? [2]

- (d) Use the words **data** and **information** to explain, in one sentence, why one sheet has given two different answers. [1]

End of paper. Check that every formula starts with = and that every word inside a formula is in quotation marks.